STAT 325 - Design of Experiment Homework 3

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Question 1

P-value calculation

```
p_value <- 1 - pf(4.3063, df1 = 4, df2 = 60)
p_value
```

```
## [1] 0.003960773
```

Question 3

```
data <- data.frame(diet = c(1, 1, 1,
           2, 2, 2, 2, 2,
           3, 3, 3, 3),

WeightGain = c(8, 16, 9,
                 9, 16, 21, 11, 18,
                 15, 10, 17, 6)
)


data_2<- data.frame(
  diet = factor(c(
    rep("1", 3),
    rep("2", 5),
    rep("3", 4)
  )),
  WeightGain = c(
    8, 16, 9,
    9, 16, 21, 11, 18,
    15, 10, 17, 6
  )
)
head(data_2)
```

```
##   diet WeightGain
## 1    1          8
## 2    1         16
## 3    1          9
## 4    2          9
## 5    2         16
## 6    2         21
```

```r
str(data)
```

```
## 'data.frame':    12 obs. of  2 variables:
##  $ diet      : num  1 1 1 2 2 2 2 2 3 3 ...
##  $ WeightGain: num  8 16 9 9 16 21 11 18 15 10 ...
```

```r
head(data)
```

```
##   diet WeightGain
## 1    1          8
## 2    1         16
## 3    1          9
## 4    2          9
## 5    2         16
## 6    2         21
```

```r
#View(data)
```

```r
table(data$diet)
```

```
##
## 1 2 3
## 3 5 4
```

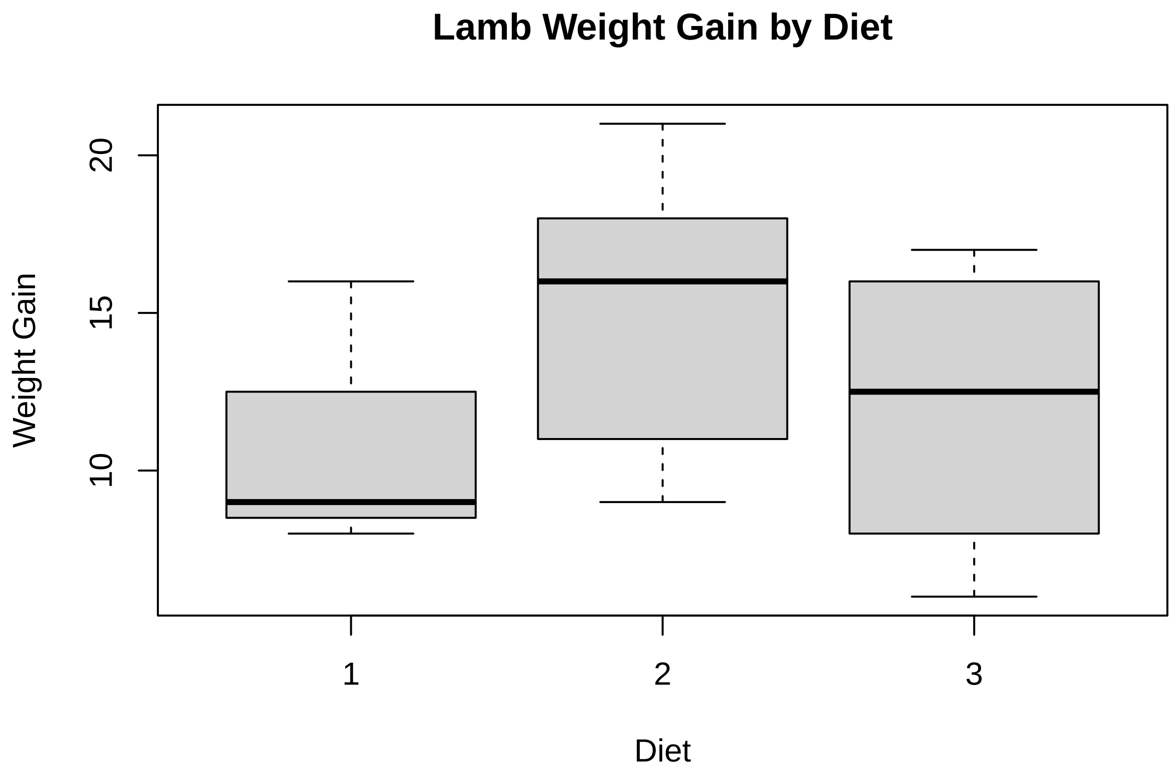
```r
summary(data)
```

```
##       diet         WeightGain
##  Min.   :1.000   Min.   : 6.00
##  1st Qu.:1.750   1st Qu.: 9.00
##  Median :2.000   Median :13.00
##  Mean   :2.083   Mean   :13.00
##  3rd Qu.:3.000   3rd Qu.:16.25
##  Max.   :3.000   Max.   :21.00
```

```r
tapply(data$WeightGain, data$diet, summary)
```

```
## $`1`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     8.0     8.5     9.0    11.0    12.5    16.0
##
## $`2`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       9      11      16      15      18      21
##
## $`3`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##     6.0     9.0    12.5    12.0    15.5    17.0
```

```r
boxplot(WeightGain ~ diet, data = data,
        xlab = "Diet",
        ylab = "Weight Gain",
        main = "Lamb Weight Gain by Diet")
```



How do the samples from the three diets compare?

From basic descriptive statistics and the boxplot we can see that diet 2 corresponds with the most weight gain the this data by distribution and sample mean. Also Diet 3 is almost symmetric with a mean of 12. Lastly we can notice that diet 1 corresponds to the least weight gain with lower variation than the other two groups in this data.

Are the mean weight gains of the three diets the same?

Although from basic descriptive summary statistics we can see that the mean weight gains of the three diets are not the same but that can be due to small sampling variation, we can also perform One-Way ANOVA to analyze the mean.

$$H_0 : \mu_1 = \mu_2 = \mu_3$$

$$H_a : H_0 \text{ is false}$$

From the ANOVA table we can see that the P value $Pr(> F) = 0.491 >> \alpha = 0.05$ *, this P value is greater than any statistically significant level, thus we can say that there is no statistical evidence to reject* H_0*. In this context, there is insufficient statistical evidence to conclude that the mean weights gains in goats differ among the three diets.*

```r
anova_model <- aov(WeightGain ~ diet, data = data_2)
summary(anova_model)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## diet         2     36   18.00   0.771  0.491
## Residuals    9    210   23.33
```

If the means are not the same, which ones are different? Since the one-way ANOVA failed to reject the null hypothesis, there is no statistical evidence that the mean weight gains differ among the three diets

Have the assumptions of the one-way ANOVA been met?

In our model assumptions for using ANOVA we have

- Independent Observations
- Errors are normally distributed
- Constant Variance

For independent observations we can assume the errors are independent across the lambs due to the design so

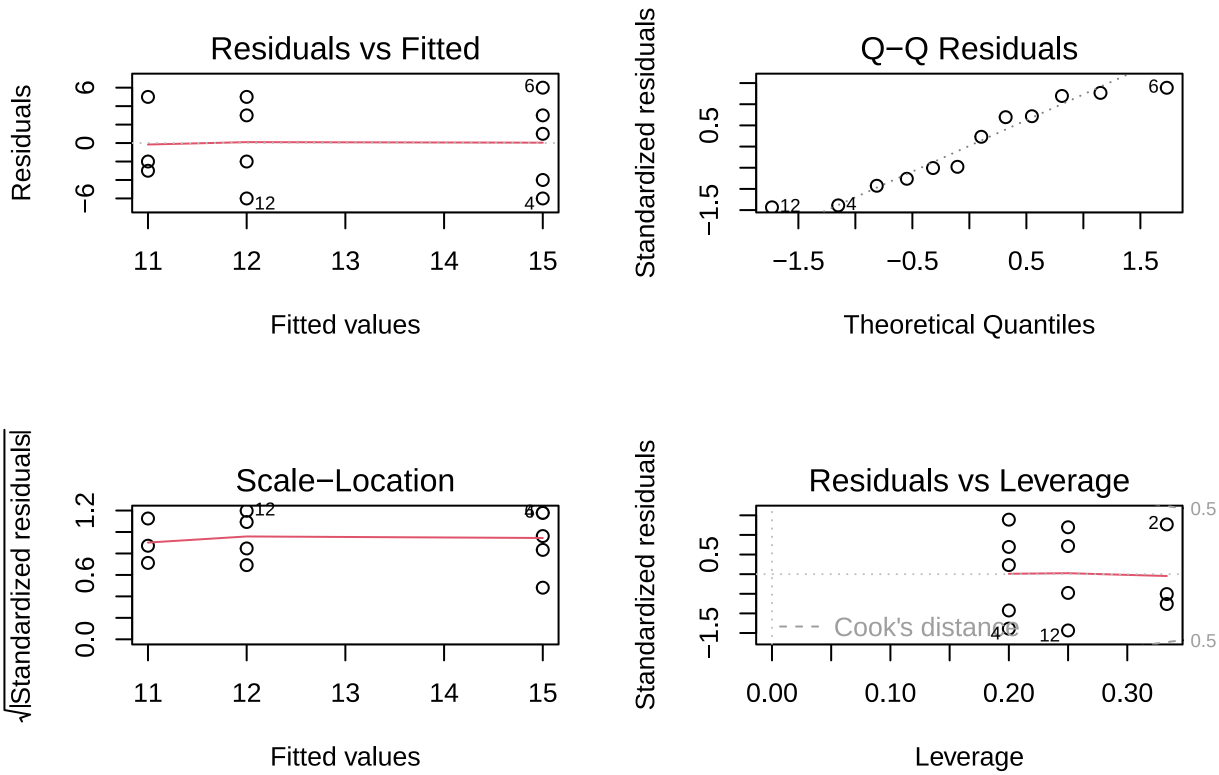
$$\varepsilon_{ij} \neq \varepsilon_{i'j'}, \quad (i,j) \neq (i',j')$$

We check normality with histogram of residuals, Shapiro-Wilk Test, and Normal probability plot/ QQ plot. With Shapiro-Wilk test we have the hypothesis as

$$H_0 : \text{Residuals are normally distributed}$$

$$H_a : \text{Residuals are not normally distributed}$$

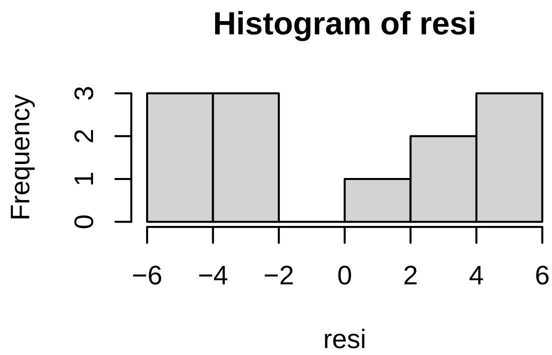
```r
par(mfrow = c(2,2))
plot(anova_model)
```




```r
resi <- residuals(anova_model)
shapiro.test(resi)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  resi
## W = 0.91275, p-value = 0.2313
```

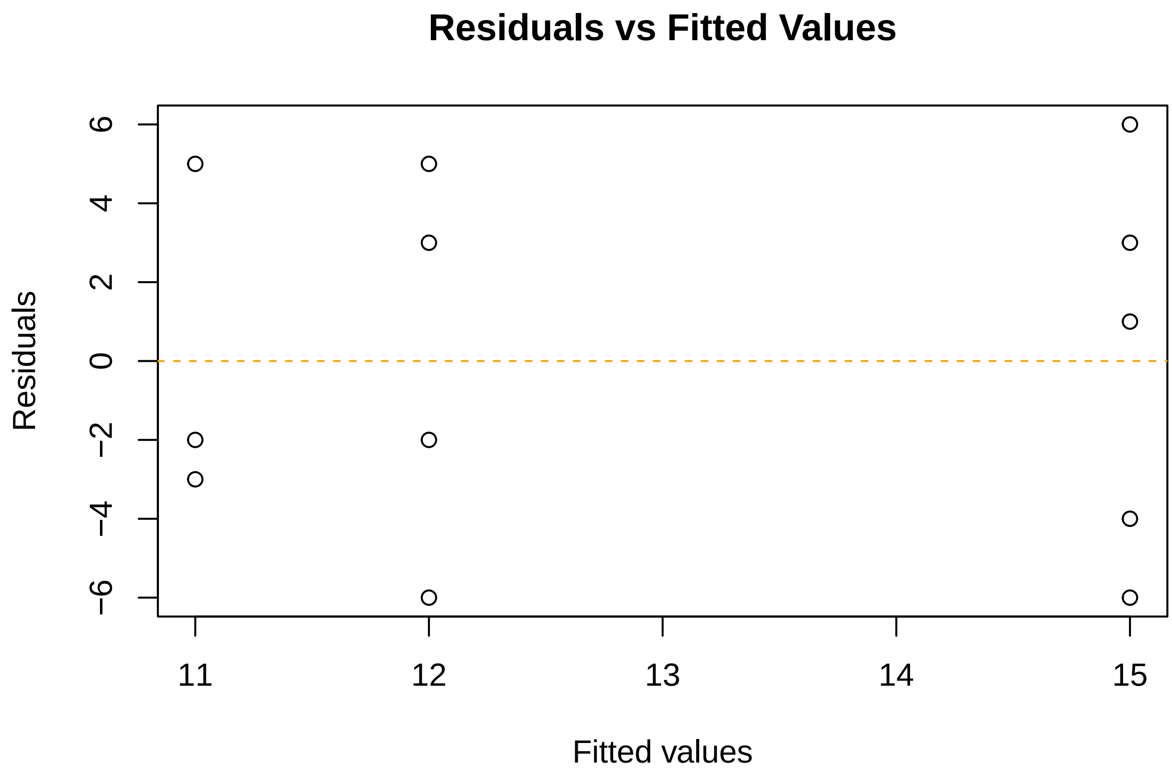
```r
hist(resi)
```



We comment that the residual histogram seems not to be normal due to small smaple size of $N = 12$ in the data, but the QQ-plot of the residuals aligns with the Gaussian distribution. Then lastly, we can see that a numeric test using Shapiro Wilk conveys that we don't have significant evidence to reject H_0 (The residuals are normally distributed). $P = 0.5394 >> \alpha = 0.05$.

Next we test constant variance assumption with looking at the residual vs fitted plot and using Bartlett's test and Levene' test.

```
plot(fitted(anova_model), residuals(anova_model),
     xlab = "Fitted values",
     ylab = "Residuals",
     main = "Residuals vs Fitted Values")
abline(h = 0, lty = 2, col = "orange")
```



The plot of the residuals seems scattered in the same fashion thus we can assume constant variance.

```r
bartlett.test(WeightGain ~ diet, data = data_2)
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  WeightGain by diet
## Bartlett's K-squared = 0.042183, df = 2, p-value = 0.9791
```

```r
library(car)
```

```
## Loading required package: carData
```

```r
leveneTest(WeightGain ~ diet, data = data_2)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##       Df F value Pr(>F)
## group  2  0.2203 0.8065
##        9
```

From both test we see that we have a large p-value which means that we fail to reject that the variances are equal. Lastly, potential outliers were examined using standardized residuals and Cook's distance, and no influential observations were identified.

```r
plot(cooks.distance(anova_model),
     type = "h",
     ylab = "Cook's Distance",
     main = "Cook's Distance")
abline(h = 4/length(residuals(anova_model)), lty = 2)
```